OmniClone: A Robust, All-Around Whole-Body Humanoid Teleoperation System

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***Abstract*—Whole-body teleoperation lets a person drive a humanoid robot through everyday tasks and collect demonstrations for autonomous learning. These tasks range from agile locomotion to precise manipulation at different body heights. Tracking policies, however, are usually judged by aggregate metrics that hide skill-specific failures, and deployment must cope with varied operators and unreliable communication. Whether one policy can cover this range on a modest training budget has remained unclear. Here we show that skill-level diagnostics can guide a single tracking policy, trained on one GPU, to cover these skills. We introduce *OmniBench*, which stratifies unseen motions by skill and difficulty, and *OmniClone*, a teleoperation system that pairs the resulting Transformer policy with initial-frame scale calibration and buffered reference delivery. Trained on 30 hours of motion data in about 82 GPU-hours on one RTX 4090, the policy lowers mean per-joint position error (MPJPE) on *OmniBench* by 65% and 78% relative to GMT and TWIST2 and reaches 39.8 mm against 38.9 mm for SONIC, which reports about 260 times as many GPU-hours. The same policy serves six operators between 1.47 and 1.94 m tall, plays back generated motions, and executes pick-and-place under a GR00T N1.6 vision-language-action (VLA) model post-trained on 50 demonstrations collected with *OmniClone*. More broadly, evaluation resolved by skill turns tracking failures into concrete choices of data and design, which makes *OmniClone* an accessible platform for humanoid control and learning from demonstrations.**

***Index Terms*—Telerobotics and Teleoperation, Humanoid Robot Systems, Humanoid and Bipedal Locomotion, Engineering for Robotic Systems.**

I. Introduction

Whole-body teleoperation lets humans guide humanoid robots through practical tasks and collect demonstrations for autonomous learning [1–6]. These tasks couple locomotion and manipulation, because reaching objects at different heights requires the whole body to move while keeping balance. Recent tracking policies have broadened the range of motions a humanoid can follow [7–11], and the broadest of them relies on large-scale data and compute [11]. The challenge is to reach broad skill coverage in a practical system whose tracking policy needs only modest data and compute.

Skill coverage is hard to assess from aggregate success rate (SR) and tracking error. A controller may track upright motions accurately yet fail during a deep squat, and an average over all motions hides this failure. Recent benchmarks grade motions by intrinsic difficulty [12] and score tracking by perceptually important artifacts [13]. For teleoperation, evaluation must also show how performance changes within a skill as workspace height or dynamic intensity varies, because such changes point to specific choices of training data and policy design.

Deployment raises further demands. Motion capture (MoCap) data must be retargeted to the robot's morphology [14], and the references must reach the controller at a steady rate. An error in the estimated operator scale distorts every retargeted trajectory, while irregular updates and interruptions break the continuity of commands. Calibration and communication therefore matter alongside tracking accuracy.

We introduce ***OmniClone***, a whole-body teleoperation system developed through diagnostic evaluation and deployment analysis. Its benchmark, ***OmniBench***, covers six skills (loco-manipulation, stationary manipulation, squatting, walking, running, and jumping), each stratified into three levels of workspace height or dynamic intensity. The reference motions are generated with HY-Motion-1.0 [15] and held out from training. The resulting diagnostics expose skill imbalances and guide controlled ablations of the motion mixture and the policy backbone.

In-place manipulation dominates the training set, yet these ablations show that shifting the rest of the mixture by a few percentage points changes dynamic tracking and manipulation stability (Sec. V-B). We refine the mixture with these diagnostics and with hardware observations, train a Transformer teacher, and distill it into a Transformer student for deployment. At deployment, initial-frame scale calibration aligns each operator's motion with the robot's nominal scale, and buffered reference delivery releases references at a fixed rate and holds the last one during interruptions.

The policy is trained on 30 hours of motion data in about 82 GPU-hours on a single RTX 4090. On ***OmniBench*** it reaches a mean SR of 97.8% and reduces mean per-joint position error (MPJPE) by 65% and 78% relative to GMT [9] and TWIST2 [16]. Its mean MPJPE of 39.8 mm is close to the 38.9 mm of SONIC [11], with lower error at every stationary manipulation and squatting level, although SONIC tracks walking more accurately and succeeds more often on high jumps (95% against 75%).

The same policy serves as the controller for generated-motion playback, real-time teleoperation, and autonomous execution by a vision-language-action (VLA) model. On hardware the robot sprints, performs long jumps, transports objects, and manipulates deformable objects (Fig. 1). Six operators between 1.47 and 1.94 m tall each complete a task that combines walking and object placement. A GR00T N1.6 model [17] post-trained on 50 demonstrations collected with ***OmniClone*** succeeds in 18 of 21 in-place pick-and-place trials and 4 of 5 squatting trials.

This paper contributes (i) ***OmniBench***, a diagnostic benchmark that resolves skill coverage by stratifying motion de-

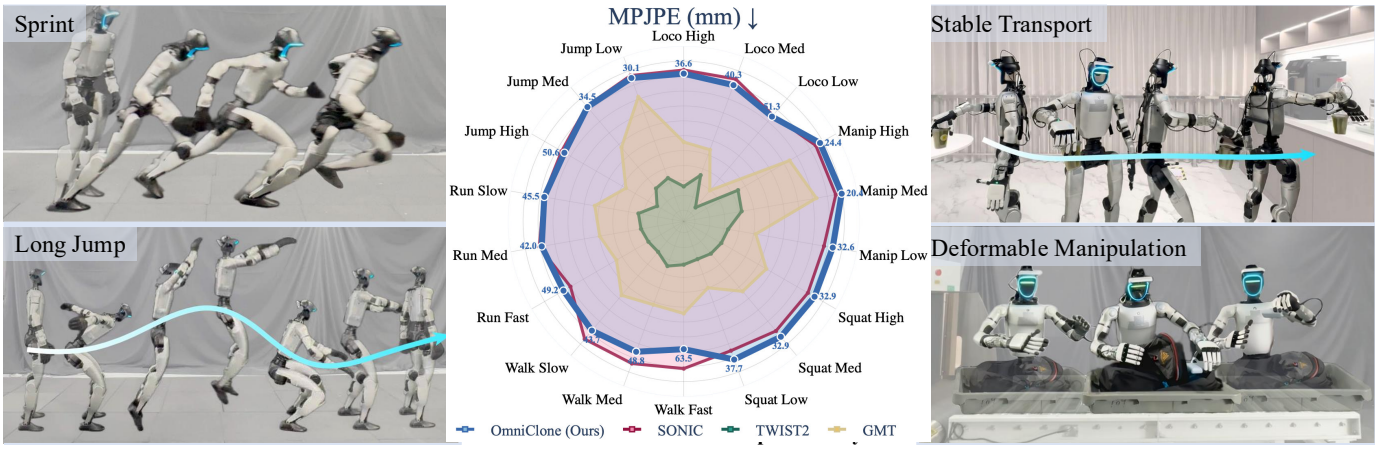


Fig. 1: **One tracking policy across skills and motion sources.** The central radar plot shows MPJPE (mm) in simulation on ***OmniBench*** for six skills at three levels each, with the radial axis inverted so that a larger extent means lower error. Loco and Manip denote loco-manipulation and stationary manipulation. ***OmniClone*** has lower error than GMT and TWIST2 in every group, and than SONIC at every stationary manipulation and squatting level. On hardware, the same policy plays back generated sprints and long jumps (left) and tracks real-time teleoperation for object transport and deformable-object manipulation (right).

mands; (ii) ***OmniClone***, a system that combines diagnosis-guided data curation, Transformer-based tracking, automatic scale calibration, and buffered reference delivery; and (iii) simulation and hardware evidence that one policy trained on a modest budget tracks a broad range of skills, works across operators and motion sources, and supports the collection of demonstrations for downstream autonomy.

II. Related Work

A. Motion Data and Diagnostic Evaluation

Whole-body imitation draws on reusable MoCap corpora such as AMASS [18] and LAFAN1 [19], while teleoperation systems collect deployment-specific motions and demonstrations [8, 16]. Recent studies weigh motion composition alongside corpus size [20, 21]. Agility Meets Stability [22] combines captured motions with synthetic balance references, whereas LIMMT [23] selects compact subsets by physical feasibility, diversity, and complexity and outperforms full-corpus training in the evaluated settings. These findings motivate treating data curation as part of controller design.

Judging these data choices requires diagnostic evaluation. Meng et al. [12] quantify the intrinsic difficulty of a motion from perturbation-induced torque variations, independently of policy performance. HumanTracker [13] provides about 150 hours of optical motion data across four motion families and introduces HumanScore, a preference-aligned metric for contact and stability artifacts. ***OmniBench*** complements these efforts with generated references in six teleoperation-relevant skills, each at three levels of workspace height or dynamic intensity. Using SR and MPJPE, it measures how performance changes within each skill as these demands grow, exposing the gaps that data curation should address.

B. Whole-Body Motion Tracking and Teleoperation

Trackers trained by reinforcement learning acquire coordinated control from human motion references. OmniH2O [24] and TWIST [25] connect tracking to teleoperation, while ExBody2 [26], GMT [9], and UniTracker [27] pursue general tracking across diverse motions. SONIC [11] broadens capabilities by scaling model, data, and compute, and HoloMotion-1 [28] combines captured and video-reconstructed motions with a mixture-of-experts Transformer and sequence-level training.

Other work examines how policies are composed and designed. TeleGate [29] gates domain-specific experts and infers a motion prior from the observation history for anticipatory control. YAHMP [30] compares command representations, history, actions, actuation, and training procedures within one pipeline, which isolates the effect of each choice from differences between complete systems. We likewise run controlled ablations of the data recipe and the backbone and use them to develop a unified Transformer teacher–student tracker under a modest training budget.

C. Motion Retargeting and Deployment

Retargeting maps human motion to the morphology and constraints of a robot. General Motion Retargeting (GMR) [14] provides a general pipeline, and Unified Motion Retargeting [31] learns dense point-cloud correspondences across motion sources and embodiments, reducing reliance on manually specified human–robot mappings.

Live deployment also requires references that are estimated and delivered consistently. HOMIE [7] pairs learned body control with an isomorphic exoskeleton cockpit, CLONE [8] limits drift through closed-loop position feedback, and TWIST2 [16] supports portable capture and demonstration collection. MOSAIC [32] corrects interface-induced errors by rapid residual adaptation, distilling interface-specific policies into an additive residual module for a general tracker. UniAct [33] streams motions generated from multimodal instructions, and SONIC [11] shares one control interface across teleoperation, generated motions, and VLA execution. We combine initial-frame scale calibration and buffered reference delivery with a policy refined through skill-level diagnostics and evaluate the resulting system across operators, motion sources, and autonomous tasks.

TABLE I: **Motion statistics of *OmniBench*.** (a) Root speed of walking and running, given as the smallest 5th percentile (Min P5), the largest 95th percentile (Max P95), and the mean of each level, with percentiles excluding start and stop transitions. (b) Root height of jumping and squatting and (c) hand height of loco-manipulation and stationary manipulation, given as the minimum, maximum, and mean of each level. Med, Loco, and Manip abbreviate medium, loco-manipulation, and stationary manipulation.

(a) Locomotion speed (m/s)

Motion	Min P5	Max P95	Mean
Walk Slow	0.618	1.704	1.026
Walk Med	0.793	2.193	1.364
Walk Fast	0.665	2.279	1.584
Run Slow	0.667	2.883	1.932
Run Med	0.644	4.151	2.386
Run Fast	0.685	5.046	2.956

(b) Root height (m)

Motion	Min	Max	Mean
Jump Low	0.522	1.206	0.783
Jump Med	0.466	1.482	0.783
Jump High	0.351	1.638	0.783
Squat Low	0.260	0.954	0.610
Squat Med	0.338	0.831	0.627
Squat High	0.357	0.818	0.690

(c) Hand height (m)

Motion	Min	Max	Mean
Loco Low	0.209	1.040	0.664
Loco Med	0.628	1.112	0.796
Loco High	0.626	1.470	0.929
Manip Low	0.162	1.007	0.698
Manip Med	0.661	1.103	0.849
Manip High	0.633	1.387	0.990

III. Skill-Level Evaluation with *OmniBench*

OmniBench is a diagnostic suite that evaluates whole-body tracking policies skill by skill. We describe its construction and then use it to evaluate existing systems.

A. Benchmark Construction

OmniBench tests tracking on motions held out from training. We generate its reference motions of daily activities at several levels of difficulty with HY-Motion-1.0 [15] and keep them out of the training data to prevent leakage. The motions span six skills, namely loco-manipulation, stationary manipulation, squatting, walking, running, and jumping. Loco-manipulation, stationary manipulation, and squatting are stratified into three levels of workspace height, walking and running into three speeds, and jumping into three heights, with 20 episodes of about 3 s per level. All references are retargeted to the robot with GMR [14], without filtering or manual correction, and evaluated in MuJoCo under identical simulation settings. Table I reports the speed and height statistics of each level.

B. Baseline Evaluation

Existing trackers fail unevenly across skills. We evaluate the released checkpoints of GMT [9], TWIST2 [16], and SONIC [11] on ***OmniBench*** (Figs. 1 and 2). GMT and TWIST2 succeed in nearly every episode of stationary manipulation at medium and high workspace heights, but they fail more often where the lower body must work harder. TWIST2 succeeds in only 65% of low loco-manipulation episodes and 70% of low squats, and GMT in 85% of low squats. Their tracking errors also exceed 100 mm in most groups, against at most 60 mm for SONIC (Tab. II). An aggregate score would hide these skill-specific gaps, which guide the data and policy choices of ***OmniClone***.

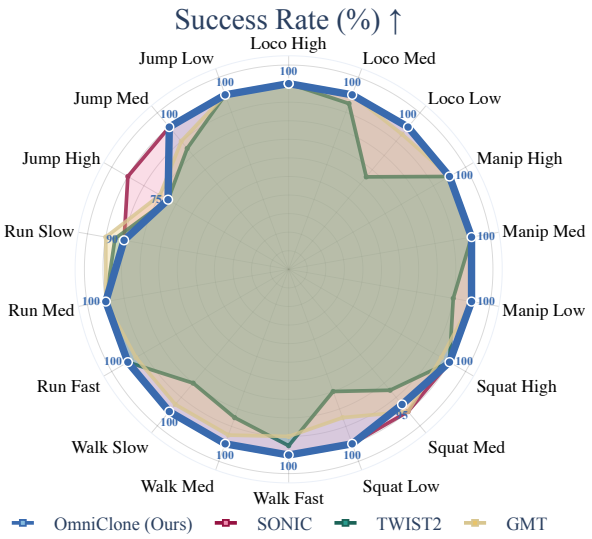


Fig. 2: **SR on *OmniBench* by skill and level.** Each axis is one of the 18 groups, and the numbers give the SR of ***OmniClone***, which succeeds in every episode in 15 groups. High jumps remain difficult for all methods, with SR of 95% for SONIC, 80% for GMT, and 75% for TWIST2 and ***OmniClone***.

IV. The *OmniClone* Teleoperation System

OmniClone pairs a whole-body tracking policy with scale calibration and buffered reference delivery (Fig. 3). Diagnostics from ***OmniBench*** shape the training data and the policy design, and the deployment pipeline connects real-time teleoperation, generated-motion playback, and VLA execution to the same controller.

A. Transformer-Based Whole-Body Tracking Policy

Whole-body tracking must integrate proprioception, motion references, and temporal context. Following Transformer-based humanoid control [34, 35], we use Transformer backbones for a teacher and a student policy, each of which outputs target angles for the robot's 29 body joints. The teacher learns from privileged kinematic observations, and the student is distilled from the teacher for deployment and uses only onboard proprioception and reference commands.

a) Teacher policy: The teacher observes joint and body-link states together with the reference motion,

$$\mathbf{o}_t^{\text{teacher}} = [\mathbf{q}_t, \mathbf{p}_t, \mathbf{g}_t, \boldsymbol{\omega}_t^{\text{root}}, \mathbf{a}_t^{\text{last}}, \mathbf{q}_t^{\text{ref}}, \mathbf{p}_t^{\text{ref}}], \tag{1}$$

where $\mathbf{q}_t$ holds joint positions and velocities, $\mathbf{p}_t$ holds body-link positions, orientations, and linear and angular velocities, $\mathbf{g}_t$ is the projected gravity, $\boldsymbol{\omega}_t^{\text{root}}$ is the root angular velocity,

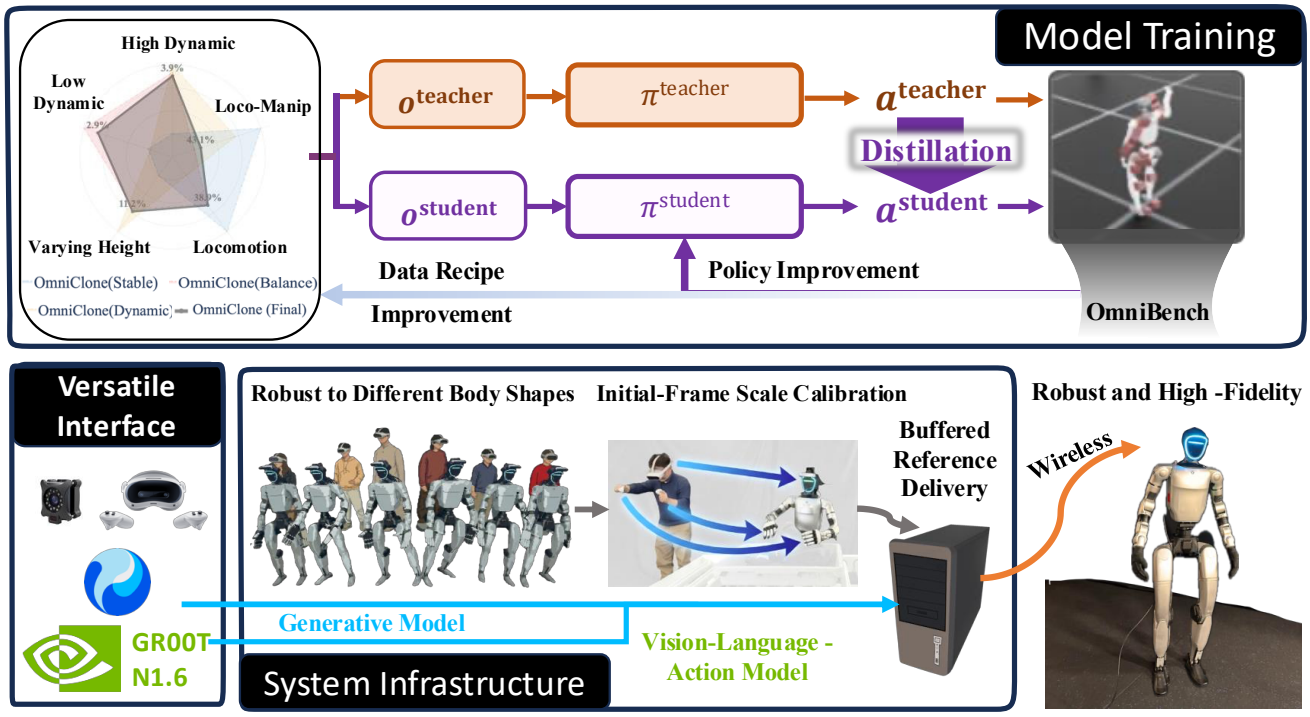


Fig. 3: **Training and deployment of *OmniClone*.** In training (top), ***OmniBench*** diagnostics refine the motion mixture and the policy design, and a Transformer teacher is distilled into a Transformer student. The motion recipes are visualized using min–max normalization. In deployment (bottom), initial-frame scale calibration aligns operator motion with the robot's nominal scale, and buffered reference delivery streams the references to the policy. Real-time teleoperation, generated-motion playback, and VLA execution all drive the same tracking policy.

$\mathbf{a}_t^{\text{last}}$ is the previous action, and $\mathbf{q}_t^{\text{ref}}$ and $\mathbf{p}_t^{\text{ref}}$ are the reference joint and body-link states.

b) Student policy: The student drops the privileged body-link states and instead receives a history of onboard proprioception,

$$\begin{aligned} \mathbf{z}_t &= [\mathbf{q}_t, \mathbf{g}_t, \boldsymbol{\omega}_t^{\text{root}}, \mathbf{a}_t^{\text{last}}], \\ \mathbf{o}_t^{\text{student}} &= [\mathbf{z}_{t-H+1}, \ldots, \mathbf{z}_t, \mathbf{c}_t^{\text{ref}}], \end{aligned} \tag{2}$$

where $\mathbf{z}_t$ is the proprioceptive observation at time t and $H = 25$ consecutive observations supply the motion history. The reference command $\mathbf{c}_t^{\text{ref}}$ contains the positions and orientations of key body links over the next five reference frames, together with the reference root linear velocity.

c) Network size: The teacher is a two-layer Transformer with an embedding dimension of 256, four attention heads, and a feed-forward dimension of 512. The student also uses four attention heads but larger embedding and feed-forward dimensions of 512 and 1024, because it must infer from its proprioceptive history what the privileged teacher observes directly. In both policies, actor and critic heads, each a multi-layer perceptron (MLP) with one hidden layer of 128 units, follow the Transformer features.

B. Training Data and Implementation

We curate a 30-hour motion corpus that spans in-place manipulation, locomotion, loco-manipulation, low-dynamic, and high-dynamic behaviors and refine its mixture with skill-level diagnostics from ***OmniBench*** and with tracking behavior observed on hardware. The corpus builds on TWIST2 [16], whose retargeted motions we expand from 23 to 29 degrees of freedom (DoFs), and adds motions captured with our own teleoperation interface. Heuristic filters on root position and velocity and on whole-body joint energy select the reference motions. Figure 3 (top left) and Tab. VII give the composition of the recipe variants compared in Sec. V-B, and Sec. A4 details data preparation. The reward combines tracking terms for torso, whole-body, and end-effector motion with regularization penalties (Tab. V), and training randomizes actuation, mass, friction, and joint-dynamics parameters (Tab. VI). We train the policy in Isaac Lab [36] on a single NVIDIA RTX 4090 with 4,096 parallel environments. Teacher training runs 33,000 Proximal Policy Optimization (PPO) steps in about 60 hours, and student distillation with DAgger takes about 22 hours under the same configuration, for a total of about 82 GPU-hours. Table III compares this budget with those reported for GMT and SONIC. SONIC reports about 260 times as many GPU-hours and 20 times as much motion data. The budget covers only the tracking policy and excludes the separate VLA post-training (Sec. VI-B).

C. Teleoperation Infrastructure

The reference pipeline addresses two deployment issues, namely scale mismatch in retargeting and irregular reference delivery.

a) Scale calibration: Differences in estimated operator scale across MoCap configurations and individuals can shift Cartesian references by up to about 20 cm. In controlled simulation tests, a ± 20 cm positional discrepancy raises MPJPE

TABLE II: **Tracking accuracy and ablations on *OmniBench*.** SR (%) and MPJPE (mm) for six skills at three levels each. Balance, Stable, and Dynamic are data-recipe variants, and MLP replaces the Transformer backbone. Bold marks the lowest MPJPE in each group.

(a) Skills stratified by workspace height

Method	Loco-Manip						Manip						Squat					
	High		Medium		Low		High		Medium		Low		High		Medium		Low	
	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓
GMT [9]	100	128.4	100	132.5	95	180.5	100	71.4	100	54.7	100	137.0	95	107.4	100	112.4	85	140.1
TWIST2 [16]	100	188.0	95	168.9	65	210.5	100	151.1	100	156.3	90	174.6	100	176.9	85	178.5	70	181.1
SONIC [11]	100	**31.7**	100	**31.6**	100	49.2	100	31.0	100	28.4	100	44.4	100	43.1	100	43.0	100	50.3
OmniClone^{Balance}	100	35.3	100	39.9	100	**48.1**	100	**19.0**	100	**18.4**	100	33.3	100	**32.0**	100	**31.4**	90	**37.1**
OmniClone^{Stable}	100	37.2	95	43.7	100	50.2	100	22.2	100	20.8	100	38.4	100	39.7	100	38.1	95	45.2
OmniClone^{Dynamic}	30	78.1	20	96.8	10	92.8	75	53.9	75	53.7	25	75.4	35	72.9	45	65.0	35	71.6
OmniClone^{MLP}	85	58.9	65	75.4	65	73.4	100	36.3	100	37.1	100	47.9	100	49.5	100	47.3	95	54.7
OmniClone	100	36.6	100	40.3	100	51.3	100	24.4	100	20.4	100	**32.6**	100	32.9	95	32.9	100	37.7

(b) Walking and running stratified by speed, jumping by height

Method	Walk						Run						Jump					
	Fast		Medium		Slow		Fast		Medium		Slow		High		Medium		Low	
	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓
GMT [9]	90	111.4	95	116.0	95	105.5	95	130.9	100	120.8	100	114.1	80	145.6	90	105.3	100	57.1
TWIST2 [16]	95	176.9	85	171.2	80	181.6	100	180.4	100	176.9	95	173.4	75	193.8	85	177.2	100	173.0
SONIC [11]	100	**37.4**	100	**32.0**	100	**28.8**	100	59.9	100	38.7	100	44.4	95	**46.3**	100	**32.9**	100	**26.6**
OmniClone^{Balance}	100	63.0	95	46.4	95	40.8	90	**45.7**	100	**38.3**	100	**41.6**	90	50.0	100	35.3	100	29.8
OmniClone^{Stable}	100	70.3	100	49.3	100	43.8	100	51.4	100	42.4	95	43.6	80	57.1	100	44.5	100	39.6
OmniClone^{Dynamic}	90	64.8	100	48.6	100	44.1	100	47.7	100	**38.3**	95	41.8	85	48.8	10	74.9	45	63.9
OmniClone^{MLP}	20	111.7	35	95.6	85	64.1	90	82.0	80	69.8	70	71.1	75	81.4	85	62.9	75	58.9
OmniClone	100	63.5	100	48.8	100	43.7	100	49.2	100	42.0	90	45.5	75	50.6	100	34.5	100	30.1

TABLE III: **Tracking-policy training resources.** Budgets of prior methods follow the cited papers, with GMT compute estimated from its reported teacher and student training times. Motion hours count training data, and GPU-hours are not normalized across hardware.

Method	Hardware	GPU-hours	Motion (h)
GMT [9]	1× RTX 4090	∼96	33.12
SONIC [11]	128 GPUs	∼21,000	611
OmniClone	1× RTX 4090	82	∼30

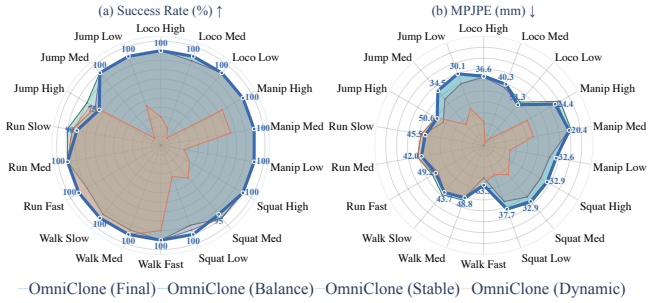


Fig. 4: **Effect of training data composition.** (a) SR and (b) MPJPE on ***OmniBench*** for the Dynamic, Stable, and Balance recipes and the final recipe used by ***OmniClone***. In (b), a larger radial extent means lower error.

from 39.8 to 69.5 mm and lowers SR from 97.8% to 85% (Tab. IX). We therefore calibrate the scale automatically from the initial frame while the operator is in an standing pose. Given the estimated operator scale h_{human} and the nominal robot scale h_{robot}, we compute $s = h_{\text{robot}}/h_{\text{human}}$ and apply it to the motion translations when constructing robot references. The factor stays fixed during teleoperation and replaces manual scale adjustment for each operator.

b) Buffered reference delivery: A data manager keeps a first-in-first-out (FIFO) queue of N_{buf} retargeted motion frames and releases references at a fixed update rate. When the stream is interrupted, it holds the last available reference. The retargeting and policy servers communicate by User Datagram Protocol (UDP) over a wireless link, so the policy can run remotely or onboard.

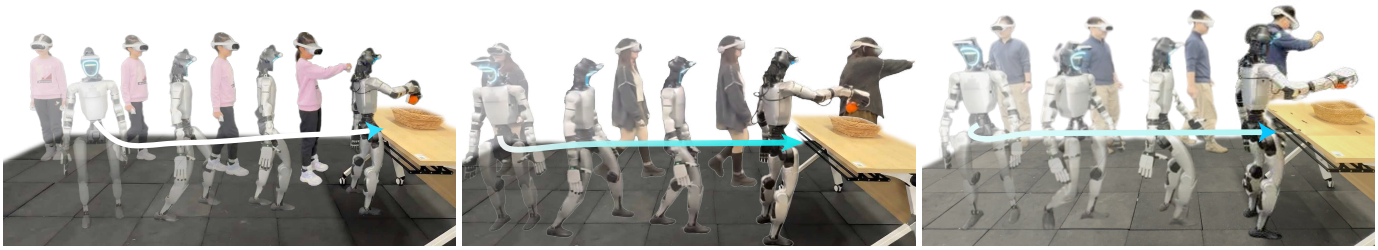
V. Simulation Experiments

We evaluate tracking in MuJoCo [37] on the 18 skill-level groups of ***OmniBench*** and report SR and MPJPE in millimeters. An episode fails when the pelvis touches the ground, failed episodes count toward MPJPE, and means weight all 18 groups equally.

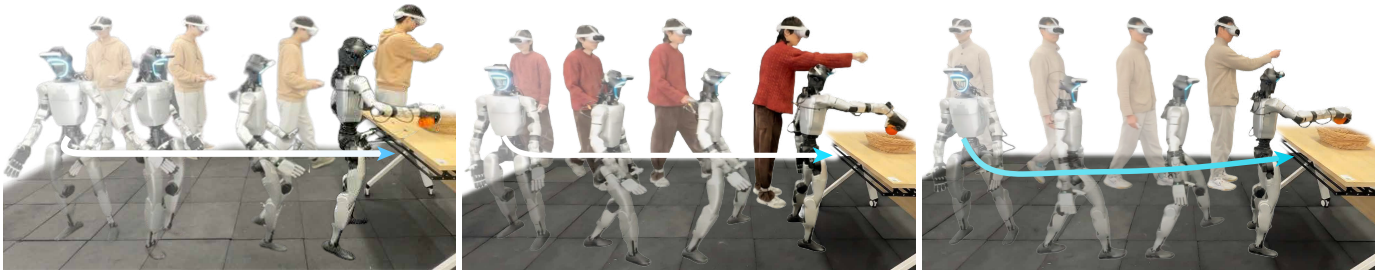
a) Baselines and variants: We compare ***OmniClone*** with the released checkpoints of GMT [9], TWIST2 [16], and SONIC [11]. Three recipe variants isolate the effect of the motion mixture (Tab. VII). ***OmniClone***^{Dynamic} slightly raises the share of highly dynamic and varying-height motions, ***OmniClone***^{Stable} that of locomotion and loco-manipulation, and ***OmniClone***^{Balance} combines the two regimes. The final recipe filters out sequences from Balance associated with tracking artifacts observed on hardware, such as jerky motion, using criteria based on root velocity and joint jitter. A parameter-matched MLP backbone, ***OmniClone***^{MLP}, is trained with the final recipe under the same settings.

A. Comparison with Baselines

On ***OmniBench***, ***OmniClone*** reaches a mean SR of 97.8% and a mean MPJPE of 39.8 mm (Tab. II). Its error is lower than that of GMT and TWIST2 in all 18 groups, and its mean MPJPE is 65% and 78% lower. Against SONIC, the mean error is similar (39.8 against 38.9 mm), and ***OmniClone*** is more accurate at every stationary manipulation and squatting level, averaging 25.8 against 34.6 mm on stationary manipulation and 34.5 against 45.5 mm on squatting. SONIC keeps a



(a) Operator A (1.47 m) (b) Operator B (1.54 m) (c) Operator C (1.65 m)



(d) Operator D (1.75 m) (e) Operator E (1.82 m) (f) Operator F (1.94 m)

Fig. 5: **Teleoperation by six operators.** Operators between 1.47 and 1.94 m tall use the same tracking policy to walk to a table, establish a stance, and pick and place an object.

higher mean SR (99.7% against 97.8%), including 95% against 75% on high jumps. ***OmniClone*** thus leads in stationary manipulation and squatting, whereas SONIC remains more accurate in walking (37.4 against 63.5 mm at fast speed) and more reliable on high jumps.

B. Ablations on Data and Backbone

a) Data recipe: Although in-place manipulation dominates the training data and the recipes differ by at most about two percentage points per category (Tab. VII), they produce distinct skill profiles (Fig. 4). The Dynamic variant succeeds in 85% of high jumps but only 10% and 45% of medium and low jumps, and it also fails often on loco-manipulation, manipulation, and squatting. Stable improves these skills and the lower jumps and reaches the highest mean SR of the three (98.1%), while Balance reaches the lowest mean MPJPE (38.1 mm). Hardware observations, however, reveal sequences in Balance associated with tracking bias, and removing them yields the final recipe. This filtering raises the mean simulation error from 38.1 to 39.8 mm while keeping the mean SR at 97.8%, so the final recipe weighs simulation diagnostics together with qualitative observations from deployment.

b) Policy backbone: Replacing the parameter-matched MLP with a Transformer lowers MPJPE in all 18 groups, from a mean of 65.4 to 39.8 mm, and raises the mean SR from 79.2% to 97.8% (Tab. II). Its SR is equal or higher in 17 of the 18 groups, and the largest gap appears in fast and medium walking, where ***OmniClone***^{MLP} succeeds in only 20% and 35% of episodes. Both variants share observations, training data, optimization settings, and training iterations, so the gain is attributable to the backbone.

VI. Real-World Experiments

We test ***OmniClone*** on hardware in real-time teleoperation, playback of generated motions, and autonomous execution after post-training on collected demonstrations. In all three settings, the same whole-body tracking policy serves as the low-level controller. The robot is a Unitree G1-Comp [38] whose 2-DoF actuated head carries an Intel RealSense D455 camera [39] and rotates through $\pm 50^\circ$ in yaw and from -90° to $+22.7^\circ$ in pitch. Operator motion from a PICO headset [40] or an optical MoCap system is retargeted on an external PC, and the tracking policy runs on the onboard computer.

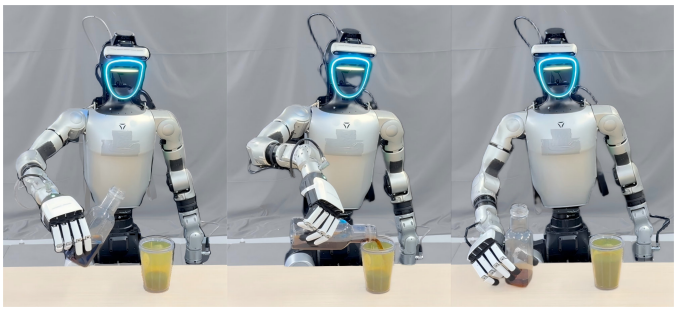
TABLE IV: **Autonomous task success after post-training.** GR00T N1.6 is post-trained on 50 demonstrations collected with ***OmniClone***, which also provides the low-level tracking policy during execution.

Task	Successes / Trials	SR (%)
Standing pick-and-place	18 / 21	85.7
Squatting pick-and-place	4 / 5	80.0

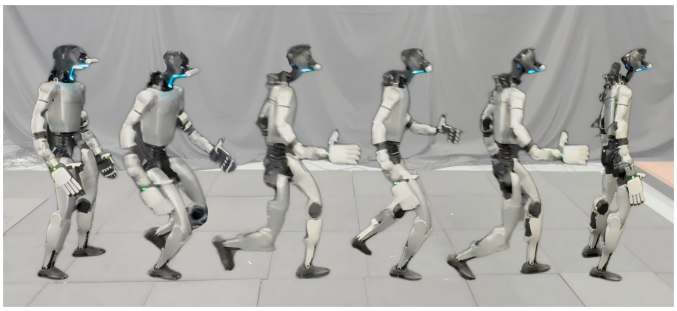
A. Teleoperation and Motion Playback

a) Real-time teleoperation: Six participants between 1.47 and 1.94 m tall teleoperate the robot to approach a table, establish a stable stance, and pick and place an object (Fig. 5). All of them complete the task with the same tracking policy and no per-operator tuning or retraining beyond the automatic scale calibration, despite large differences in body height and proportions. Most participants had never teleoperated a robot, and every novice succeeded within five to seven practice trials. These results indicate that ***OmniClone*** accommodates operators of widely different builds and that new operators learn to use it quickly. The supplementary video shows further executions.

We further teleoperate tasks that combine manipulation with changes in posture and locomotion, such as transporting coffee beans (Fig. 1), pouring tea (Fig. 6a), and opening a low cabinet to retrieve a tissue box (Fig. 7). The robot also runs under teleoperation (Fig. 6b). One policy thus supports precise manipulation, composite loco-manipulation, and dynamic locomotion without switching controllers.



(a) Pouring tea

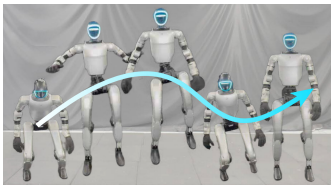


(b) Running

Fig. 6: **Manipulation and running under one tracking policy.** Real-time teleoperation of (a) tea pouring and (b) running, two tasks that span precise liquid handling and dynamic locomotion.



Fig. 7: **Teleoperation of a composite retrieval task.** The robot opens a low cabinet, squats to retrieve a tissue box, walks to a table, and places the box on it, with one tracking policy coordinating posture changes, locomotion, and manipulation.



(a) Jumping in place



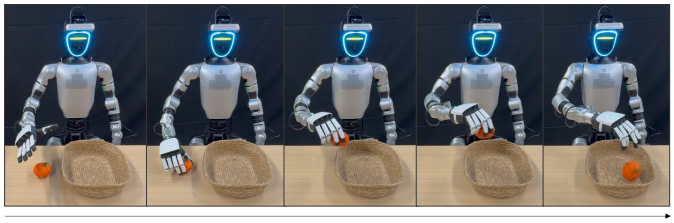
(b) Stretching and punching

Fig. 8: **Hardware playback of generated motions.** Motions generated by HY-Motion-1.0 run on the same tracking policy without further training, here (a) a jump in place and (b) a stretching and punching sequence.

b) Generated-motion playback: We generate reference motions outside the training corpus with HY-Motion-1.0 [15] and execute them on hardware without further policy training. They include sprinting and long jumping (Fig. 1) as well as jumping in place, stretching, and punching (Fig. 8). Generated references thus drive the same controller as live teleoperation, which suggests that ***OmniClone*** can serve as a low-level controller beneath high-level motion planners.

B. Data Utility for Autonomy

We post-train GR00T N1.6 [17], a VLA model, on 50 whole-body demonstrations collected with ***OmniClone*** and evaluate two tasks. In standing pick-and-place, the robot lifts a tomato and places it in a basket (Fig. 9a), and in squatting pick-and-place, it performs the same manipulation from a squat (Fig. 9b). During autonomous execution, GR00T predicts commands for the same tracking policy that was used to collect the demonstrations, so the test covers the full path from teleoperated data collection to autonomous control. Post-training runs for about 20,000 steps on four NVIDIA A100 GPUs with the vision-language backbone frozen. At inference, GR00T takes the head-camera image, the language instruction, and the proprioceptive state and predicts a 16-step action chunk, of which ***OmniClone*** executes the first eight before



(a) Standing pick-and-place.



(b) Squatting pick-and-place.

Fig. 9: **Autonomous pick-and-place.** GR00T N1.6, post-trained on demonstrations collected with ***OmniClone***, picks up a tomato and places it in a basket with (a) standing and (b) squatting poses. ***OmniClone*** provides low-level whole-body control in both tasks.

the next query. Forward kinematics converts the joint-space outputs into body-link reference commands for the tracking policy, a step that can add noise to the references.

The autonomous policy succeeds in 18 of 21 standing trials and 4 of 5 squatting trials (Tab. IV). Demonstrations collected with ***OmniClone*** thus support learning both evaluated tasks, and the same tracking controller carries over from human-driven to autonomous operation. Section C2 details data collection and post-training.

VII. Conclusion

Skill-resolved evaluation on ***OmniBench*** exposes how the skills of a tracking policy depend on the composition of its training motions, and these diagnostics guide ***OmniClone***, a teleoperation system whose single tracking policy trains on 30 hours of motion data in about 82 GPU-hours on one RTX 4090. The same policy accepts live and generated references, serves operators between 1.47 and 1.94 m tall, and

executes pick-and-place under a VLA model post-trained on 50 collected demonstrations. Three limits bound these results. High jumps still succeed less often than with SONIC (75% against 95%), walking remains less accurate (63.5 against 37.4 mm at fast speed), and hardware results beyond the 26 autonomous trials on two tasks are qualitative. Closing the gap in dynamic locomotion, broadening the hardware and autonomous evaluation, and extending ***OmniBench*** with standardized real-world task protocols are the main next steps.

REFERENCES

[1] K. Darvish, L. Penco, J. Ramos, R. Cisneros, J. Pratt, E. Yoshida, S. Ivaldi, and D. Pucci, "Teleoperation of humanoid robots: A survey," *IEEE Transactions on Robotics (T-RO)*, vol. 39, no. 3, pp. 1706–1727, 2023.

[2] Z. Fu, Q. Zhao, Q. Wu, G. Wetzstein, and C. Finn, "HumanPlus: Humanoid shadowing and imitation from humans," in *Conference on Robot Learning (CoRL)*, 2024.

[3] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi, "Learning human-to-humanoid real-time whole-body teleoperation," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[4] H. Jiang, J. Chen, Q. Bu, L. Chen, M. Shi, Y. Zhang, D. Li, C. Suo, C. Wang, Z. Peng *et al.*, "WholeBodyVLA: Towards unified latent VLA for whole-body loco-manipulation control," in *Proceedings of International Conference on Learning Representations (ICLR)*, 2026.

[5] Y. Wang, Q. Zhao, Y. F. Lau, R. Yu, H. W. Tsui, Q. Chen, J. Wang, J. Pang, and P. Tan, "HumanX: Toward agile and generalizable humanoid interaction skills from human videos," in *Conference on Robot Learning (CoRL)*, 2026.

[6] R. Dong, Z. Li, X. He, and S. Gupta, "Learning humanoid end-effector control for open-vocabulary visual loco-manipulation," in *Conference on Robot Learning (CoRL)*, 2026.

[7] Q. Ben, F. Jia, J. Zeng, J. Dong, D. Lin, and J. Pang, "HOMIE: Humanoid loco-manipulation with isomorphic exoskeleton cockpit," in *Robotics: Science and Systems (RSS)*, 2025.

[8] Y. Li, Y. Lin, J. Cui, T. Liu, W. Liang, Y. Zhu, and S. Huang, "CLONE: Closed-loop whole-body humanoid teleoperation for long-horizon tasks," in *Conference on Robot Learning (CoRL)*, 2025.

[9] Z. Chen, M. Ji, X. Cheng, X. Peng, X. B. Peng, and X. Wang, "GMT: General motion tracking for humanoid whole-body control," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[10] Z. Zhang, C. Chen, H. Xue, J. Wang, S. Liang, Y. Liu, Z. Zhang, H. Wang, and L. Yi, "Unleashing humanoid reaching potential via real-world-ready skill space," *IEEE Robotics and Automation Letters (RA-L)*, vol. 11, no. 2, pp. 2082–2089, 2026.

[11] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben *et al.*, "SONIC: Supersizing motion tracking for natural humanoid whole-body control," *Science Robotics*, vol. 11, no. 117, p. eaed4592, 2026.

[12] Z. Meng, L. Yin, X. Chen, C. Zuo, A. Chen, S. Guo, and Y. Qin, "Distinguishing imitation error from intrinsic motion learning difficulty," in *Proceedings of International Conference on Machine Learning (ICML)*, 2026.

[13] D. Liu, Z. Qi, J. Zeng, Y. Guan, C. Lin, X. Chen, X. Yu, W. Zhang, H. Wang, and L. Yi, "HumanTracker: Towards comprehensive and human-aligned motion tracking benchmark," in *Proceedings of European Conference on Computer Vision (ECCV)*, 2026.

[14] J. P. Araujo, Y. Ze, P. Xu, J. Wu, and C. K. Liu, "Retargeting matters: General motion retargeting for humanoid motion tracking," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[15] Tencent Hunyuan 3D Digital Human Team, "HY-Motion 1.0: Scaling flow matching models for text-to-motion generation," *arXiv preprint arXiv:2512.23464*, 2025.

[16] Y. Ze, S. Zhao, W. Wang, A. Kanazawa, R. Duan, P. Abbeel, G. Shi, J. Wu, and C. K. Liu, "TWIST2: Scalable, portable, and holistic humanoid data collection system," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[17] NVIDIA GEAR Team, "GR00T N1.6: An improved open foundation model for generalist humanoid robots," https://research.nvidia.com/labs/gear/gr00t-n1_6/, Dec. 2025.

[18] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, "AMASS: Archive of motion capture as surface shapes," in *Proceedings of Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[19] F. G. Harvey, M. Yurick, D. Nowrouzezahrai, and C. Pal, "Robust motion in-betweening," in *ACM SIGGRAPH / Eurographics Symposium on Computer Animation (SCA)*, 2020.

[20] J. Li, X. Cheng, T. Huang, S. Yang, R.-Z. Qiu, and X. Wang, "AMO: Adaptive motion optimization for hyper-dexterous humanoid whole-body control," in *Robotics: Science and Systems (RSS)*, 2025.

[21] W. Xie, J. Han, J. Zheng, H. Li, X. Liu, J. Shi, W. Zhang, C. Bai, and X. Li, "KungfuBot: Physics-based humanoid whole-body control for learning highly-dynamic skills," in *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[22] Y. Pan, R. Qiao, L. Chen, K. Chitta, L. Pan, H. Mai, Q. Bu, H. Zhao, C. Zheng, P. Luo *et al.*, "Agility meets stability: Versatile humanoid control with heterogeneous data," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[23] Y. Guan, Z. Qi, C. Lin, X. Chen, D. Liu, W. Zhang, J. Wang, X. Yu, H. Wang, and L. Yi, "LIMMT: Less is more for motion tracking," in *Proceedings of International Conference on Machine Learning (ICML)*, 2026.

[24] T. He, Z. Luo, X. He, W. Xiao, C. Zhang, W. Zhang, K. Kitani, C. Liu, and G. Shi, "OmniH2O: Universal and dexterous human-to-humanoid whole-body teleoperation and learning," in *Conference on Robot Learning (CoRL)*, 2024.

[25] Y. Ze, Z. Chen, J. P. Araújo, Z.-a. Cao, X. B. Peng, J. Wu, and C. K. Liu, "TWIST: Teleoperated whole-body imitation system," in *Conference on Robot Learning (CoRL)*, 2025.

[26] M. Ji, X. Peng, F. Liu, J. Li, G. Yang, X. Cheng, and X. Wang, "ExBody2: Advanced expressive humanoid whole-body control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[27] K. Yin, W. Zeng, K. Fan, M. Dai, Z. Wang, Q. Zhang, Z. Tian, J. Wang, J. Pang, and W. Zhang, "UniTracker: Learning universal whole-body motion tracker for humanoid robots," *IEEE Robotics and Automation Letters (RA-L)*, vol. 11, no. 7, pp. 8124–8131, 2026.

[28] M. Chen, K. Wang, B. Zhang, X. Ma, Z. Yang, Y. Ren, Q. Huang, Z. Zhu, Y. Wang, and Z. Su, "HoloMotion-1 technical report," *arXiv preprint arXiv:2605.15336*, 2026.

[29] J. Li, B. Tang, and F. Wu, "TeleGate: Whole-body humanoid teleoperation via gated expert selection with motion prior," in *Robotics: Science and Systems (RSS)*, 2026.

[30] F. Amadio and E. Mingo Hoffman, "What matters in humanoid general motion tracking? An empirical study," *arXiv preprint arXiv:2607.19903*, 2026.

[31] H. Cao, Y. Fang, T. Kwon, R. Yu, J. Ma, J. Tan, Y. Zhou, B. Du, Y. Gu, Y. Gao, R. Dai, L. Han, and R. Xu, "Unified motion retargeting for humanoids with learned point cloud correspondence," *arXiv preprint arXiv:2609.02134*, 2026.

[32] Z. Sun, B.-S. Huang, Y. Peng, X. Li, J. Ma, Y. Sun, Z. Li, H. Jiang, B. Gao, Z. Bing, X. Wang, and A. Knoll, "MOSAIC: Bridging the sim-to-real gap in generalist humanoid motion tracking and teleoperation with rapid residual adaptation," *arXiv preprint arXiv:2602.08594*, 2026.

[33] N. Jiang, Z. He, W. Yu, L. Pang, Y. Li, H. Li, J. Cui, Y. Li, Y. Wang, Y. Zhu *et al.*, "UniAct: Unified motion generation and action streaming for humanoid robots," *arXiv preprint arXiv:2512.24321*, 2025.

[34] J. He, C. Zhang, F. Jenelten, R. Grandia, M. Bächer, and M. Hutter, "Attention-based map encoding for learning generalized legged locomotion," *Science Robotics*, vol. 10, no. 105, p. eadv3604, 2025.

[35] I. Radosavovic, T. Xiao, B. Zhang, T. Darrell, J. Malik, and K. Sreenath, "Real-world humanoid locomotion with reinforcement learning," *Science Robotics*, vol. 9, no. 89, p. eadi9579, 2024.

[36] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrügg, N. Rudin *et al.*, "Isaac Lab: A GPU-accelerated simulation framework for multi-modal robot learning," *arXiv preprint arXiv:2511.04831*, 2025.

[37] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.

[38] Unitree Robotics, "Unitree G1-Comp," https://www.unitree.com/cn/robocup, 2025.

[39] Intel Corporation, "Intel® RealSense™ D455," https://www.intel.com/content/www/us/en/products/sku/205847/intel-realsense-depth-camera-d455/specifications.html, 2020.

[40] PICO, "PICO 4 Ultra," https://www.picoxr.com/cn/products/pico4-ultra, 2025.